

“In future homes, all household technology will be networked”

The Internet of Things (IoT) is growing fast with every design house, worth its salt, building devices using its paradigm. As a consequence, things have gotten extremely competitive (and frustrating) for design houses. That is why EFY decided to discuss with an industry veteran on how to get noticed with your next IoT product. Abhishek Mutha speaks to R.K. Shenoy, senior vice president, Powertrain Electronics, Robert Bosch Engineering and Business Solutions Limited, to source some advice



R.K. SHENOY
SR. VICE PRESIDENT, POWERTRAIN ELECTRONICS,
ROBERT BOSCH ENGINEERING AND BUSINESS
SOLUTIONS LIMITED

What is the biggest factor that, if implemented, could affect the evolution of IoT?

The advent of nanotechnology will enable smarter sensors and actuators that can also capture context information along with data. Having tiny wireless sensors that operate on ultra-low power to be able to work for 10-20 years, and communicate wirelessly, is one key element to achieve and Bosch is active on this. Further, there will be both active and passive devices getting connected to the Internet. Technologies like IPv6 will be an enabler to manage such complex networks in a modular, scalable and configurable IoT model.

What challenges should engineers focus on, to build an incredible IoT device?

There are blocking points that need to be overcome to build incredible devices, for example, battery-less sensors. For IoT to reach its full potential, sensors will need to be self-sustaining. Imagine changing batteries in billions of devices deployed across the planet and even into space. Obviously, this is not possible. What is needed is a way for sensors to generate electricity from environmental elements such as vibrations, light and airflow. Next is the quality of service of the connectivity. When more and more critical decisions can be autonomously taken, the data connectivity has to be guaranteed. Hence the penetration of IoT will not happen overnight. The other thing is the psychological aspect of 'get going,' the decision making to machines in an M2M context.

Engineers working on what areas stand to make the most impact with the IoT?

With the advent of IoT and approximately 6.5 billion connected devices predicted by 2016, there would be a lot of advancement in all domains—right from production, energy management, security and infrastructure (smart towns). Autonomous vehicles will drive us around without the need of our attention on the road, and this is an extremely focussed area at the moment. In the telemedicine sector, embedded body sensors would revolutionise the medical industry.

Homes will be wired from the basement to the attic through home automation. In future homes, all household technology will be networked, allowing homeowners to control the entire system autonomously and monitor via an app on their tablet or smartphone. An innovation team at Bosch is working on this vision of a networked, intelligent home. In the house of tomorrow, the heating system detects when a window is open and simply turns off the heat in that room. If the weather forecast calls for rain, a warning appears on the homeowner's smartphone to inform her that a window is open. Also, the security system alarm sounds on the phone if a window is broken or smoke is detected.

Predictive maintenance and remote diagnostics in vehicles will become standard. The next logical step is preventative diagnostics: If a component is on the verge of failing, the car automatically reports this to the service centre. Bosch is working on this, as well as on software for fleet management. The latter will allow leasing firms or car-sharing op-